Computer Communications 154 (2020) 92–98

Contents lists available at ScienceDirect

Computer Communications

journal homepage: www.elsevier.com/locate/comcom

A remote laser focusing system with spatial light modulator

Lin Liu [a,b], Lin Li [a,*], Zhu Zhao [b], Yue Wang [b]

[a] Beijing Institute of Technology, Beijing, China
[b] The 11th research institute of China electronics technology group corporation, Beijing, China

ARTICLE INFO

Keywords:
Remote laser focusing (RLF)
Spatial light modulator (SLM)
Wavefront modulation
Mechanical focusing

ABSTRACT

This paper designs a remote laser focusing system (RLFS) based on spatial light modulator (SLM). In this system, the SLM serves as an active optical element that modulates the wavefront of the laser beam, causing variation in beam transmission. Theoretical analysis shows that a reflective SLM can compensate the residual aberrations at different operating distances, enabling the accurate focusing of targets without sacrificing the energy convergence. Based on optical design software, the geometrical parameters were optimized for the presented system through simulation. Finally, experimental demonstration and error analysis were carried out. The results show that the proposed RLFS-SLM is more stable and capable for controlling energy convergence than the traditional mechanical RLFS. Our system provides a desirable tool for high-energy laser applications.

1. Introduction

Lasers have been widely used in various sectors of the national economy, due to its high brightness, directivity, monochromaticity and coherence. Laser beams can be focused to very tiny spots, achieving a very high irradiance. If the energies are high enough, the beams can burn or move objects over a long distance within a short time. Therefore, many remote laser focusing systems (RLFSs) have emerged to replace humans for dangerous tasks, ranging from laser cleaning [1,2], dismantling of nuclear facilities [3,4], to defense and security [5]. The RLFSs can be used at laser cleaning, laser interception and striking, laser processing. However, the problem of rlfss is that the object distance is constantly changing for different distance targets or moving targets. Rlfss has to solve the problem of focusing spot diffusion and energy loss with the change of distance. The key to the RLFSs lies in remote laser focusing. The impacts, schemes and far-field control of remote laser focusing have long been research hotspots [6–8]. For example, Zhang et al. [9] designed a RLFS with reflective optics in its optical system, aiming to prevent lens from being damaged by the high-energy laser.

There are generally two structure forms of the focusing device in existing RLFSs, namely, fixed-focusing and auto-focusing. The fixed-focusing structure only removes obstacles at fixed distances. Obstacles placed at different distances require an auto-focusing structure, which is traditionally achieved by the axial movement of one mirror. However, the current auto-focusing method face three defects: (1) The energy is concentrated only at specific positions rather than across the operating range of the laser power; (2) The movement of one mirror theoretically causes a defocusing effect called off-axis aberration, which degrades the convergence effect; the off-axis aberration intensifies with the distance to the object; (3) The axial movement of the system requires high accuracy of structural design; otherwise, any error of the optics axis will severely suppress the focusing effect.

At present, the hot spot technology also includes adaptive optical closed-loop control technology, which uses deformable mirror to adjust. This method can achieve real-time adjustment and high precision, but the control system is complex, high precision, slow response time and high cost. Therefore, the leading challenges facing the RLFS technology are high focusing accuracy simple operation and high-energy concentration at different distances.

To cope with the challenges, this paper presents a RLFS based on a spatial light modulator (SLM). It can be widely used in optical computing, pattern recognition, information processing, display and other fields, and has broad application prospects. For convenience, the proposed system is denoted as the RLFS-SLM in the following discussions. The reflective SLM can compensate the residual aberrations by the designed surface, enabling accurate focusing on targets and ensuring the energy convergence effect. Experimental results show that the remote focusing effect of the laser is enhanced by the high-precision control of the SLM, which is more feasible than the traditional mechanical approach. The key technology to be solved in this paper is to use a simple, fast and low-cost method to realize the high-precision focusing of laser rlfss at different distances, so as to achieve the purpose of high energy concentration at different distances.

The remainder of this paper is organized as follows: Section 2 explains the optical principle of the traditional RLFS, and describes the working principles of the RLFS-SLM; Section 3 introduces the optical

* Corresponding author.
E-mail address: li_lin@263.net (L. Li).

https://doi.org/10.1016/j.comcom.2020.01.075
Received 9 December 2019; Received in revised form 17 January 2020; Accepted 30 January 2020
Available online 4 February 2020
0140-3664/© 2020 Elsevier B.V. All rights reserved.

design of the RLFS-SLM, simulates the system on optical software, and theoretically analyzes the size and intensity of the laser spot; Section 4 verifies the feasibility of the RLFS-SLM through experiments; Section 5 wraps up this research with conclusions.

2. Theoretical principles

2.1. Optical principle of the traditional RLFS

As shown in Fig. 1, the optical principle of the traditional RLFS can be described as a Galileo telescope system in reverse. First, the high-power laser device emits a collimated beam of a certain diameter. Then, the beam is expanded by the secondary mirror, and the divergence angle is compressed by the primary mirror into a focused spot, making a powerful strike on the target.

As the secondary mirror is translated from L_{m1} to L_{m2} and finally to L_{mn}, the laser beam will be focused at different positions, corresponding to L_{f1}, L_{f2}, and L_{fn} in Fig. 1, respectively. For given target ranging and position, the laser beam can be focused on the targets of different distances, simply through axial motions of the secondary mirror.

In mechanical RLFS, however, inherent errors are inevitably introduced by the manufacturing and assembling of mechanical structure and elements. The positional accuracy of the secondary mirror is also dampened by the movement error of the platform. The errors in the system cause off-axis aberrations of the reverse Galileo telescope system, pushing up the radius of the focused spot while reducing the intensity.

2.2. Working principles of the RLFS-SLM

(1) Optical principle

To improve the performance of the RLFS, the authors replaced the secondary mirror of the traditional mechanical method with an SLM (Fig. 2). When a laser beam is transmitted to the reflected SLM, it will be modulated by the designed wavefront of the SLM to focus at the required position (e.g. Z_{f1}, Z_{f2}, …, or Z_{fn}). If the designed wavefront of the SLM is sufficiently rational and precise, the system aberration will be corrected in the modulation process. This means the RLFS-SLM can theoretically outperform the traditional mechanical method.

(2) Modulation principle of the SLM

The SLM is an electronic programmable device, capable of modulating the amplitude, phase, and frequency of the refracted and reflected lights [10,11]. Liquid crystal SLMs (LC-SLMs) boast several advantages in engineering product design: simple production, wide adjustment range, cheap price, low energy consumption, and easy control [12]. The LC-SLMs can be either optical addressing or electrical addressing. The reflective optical addressable LC-SLM has a relatively high energy threshold [13,14], offering a desirable tool for high-energy laser applications.

With the aid of digital holograms, the SLMs enable the control over the generation of optical fields [11,15]. The structure of the LC-SLM is shown in Fig. 3. To correct wavefront distortion, the LC-SLM controls the optical-phase shift to compensate the optical path difference (OPD):

$$OPD = \Delta n_z \cdot z \tag{1}$$

where, Z is the thickness of the LC layer in the LC-SLM; Δn_z is the difference between the two media in refractive index. The LC exerts an electronically controlled birefringence effect, which divides the incident light into light $\boldsymbol{e}$ and light $\boldsymbol{o}$, with refractive indices of n_e and n_o, respectively. The refractive index of light $\boldsymbol{e}$ can be changed by applying a voltage to the LC layer, inducing a certain phase delay of the incident light.

Under the applied voltage V, the rotation angle of an LCD molecule can be described as:

$$\theta = \frac{\pi}{2} - 2\arctan\left(e^{-V}\right) \tag{2}$$

The applied voltage causes the LC molecules to rotate along the direction of the electric field. Then, the refractive index of light $\boldsymbol{e}$ becomes:

$$n_{e(V)} = \frac{n_e \cdot n_o}{\sqrt{n_e^2 \sin^2\theta + n_o^2 \cos^2\theta}} \tag{3}$$

When the voltage is applied to the surface of the LC layer, the incident light will be polarized in parallel to the optical axis of the LC. Then, the LC-SLM generates a pure phase modulation of the incident light:

$$\delta = 2\pi d \cdot \frac{n_{e(V)} - n_o}{\lambda} \tag{4}$$

where, δ is the phase delay; d is the thickness of the LC layer; λ is the wavelength of the incident light.

Therefore, the voltage applied to each pixel of the LC layer can be changed by setting the relative polarization azimuths of different polarizers. Then, an OPD will be generated by the altered birefringence of the incident polarized light to the LC. The phase of the incident light can be modulated by changing the OPD, yielding the modulation mode and modulation characteristic curve of the corresponding SLM.

(3) Calculations of the position and diameter of the focused spot

The main component of the RLFS is an inverse Galileo telescope system. According to the Gaussian beam transmission formulas, the position and size of the beam waist as well as the size of the focused spot at any position can be calculated based on the focal lengths and the relative position between primary and secondary mirrors. Theoretically, the size and position of the beam waist change with the focal length of the secondary mirror.

For a fiber laser, a Gaussian beam of beam waist radius ω_0 and wavelength λ is assumed to pass through a thin lens. Suppose parameter $F = 2{\omega_0}^2/[1+\exp({\omega_0}^2)]$. The position and size of the beam waist behind the lens can be respectively obtained by:

$$z_f = \frac{f}{1 + \frac{\pi^2 N_0^2}{(1+2\omega_0 - F)(1-F)}} \tag{5}$$

$$\omega_f = \frac{\omega_0}{1 + \frac{\pi^2 N_0^2}{(1+2\omega_0 - F)(1-F)}} \tag{6}$$

where, f is the focal length of the lens; ${N_0}^2 = \lambda f {\omega_0}^2$ is the Fresnel number of the Gaussian beam. Eqs. (5) and (6) show that the position and the radius of the focused spot depend on the f, which can be accurately controlled by the SLM.

3. Optical design and simulations

This paper designs and optimizes the RLFS-SLM based on five operating distances (z_{f1} to z_{f5}), and designs the primary mirror at the operating distance of z_{f3}.

3.1. Focal length design of the SLM

The focal length of the secondary mirror (SLM) was computed by Eq. (5). The parameters were set as follows: $\lambda = 1080$ nm $\omega_0 = 22$ mm, $z_{f1} = 0.5$ km, $z_{f2} = 0.75$ km, $z_{f3} = 1$ km, $z_{f4} = 1.25$ km, and $z_{f5} = 1.5$ km. The diameter of the remote focused spot was 39 mm. The focused spot was assumed to have a constant power, creating an ideal scenario for theoretical verification of the proposed optical system. Table 1 lists the focal lengths of the secondary mirror (SLM) at different operating distances.

3.2. Optical design of the system

The optical design of the system was conducted on Codev, a commercial software suite for optical design. The wavefront error was minimized through optimization of the Zernike polynomials of the

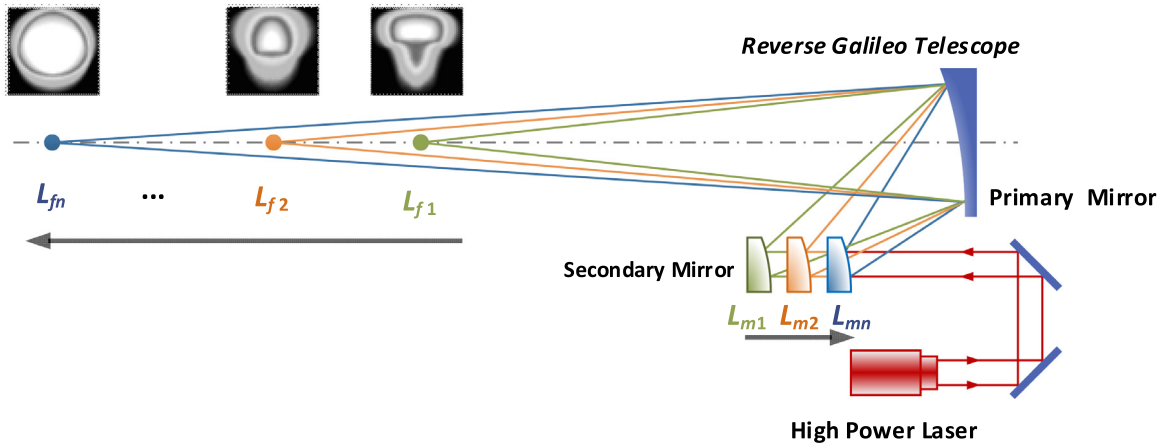


Fig. 1. Optical principle of the traditional RLFS.

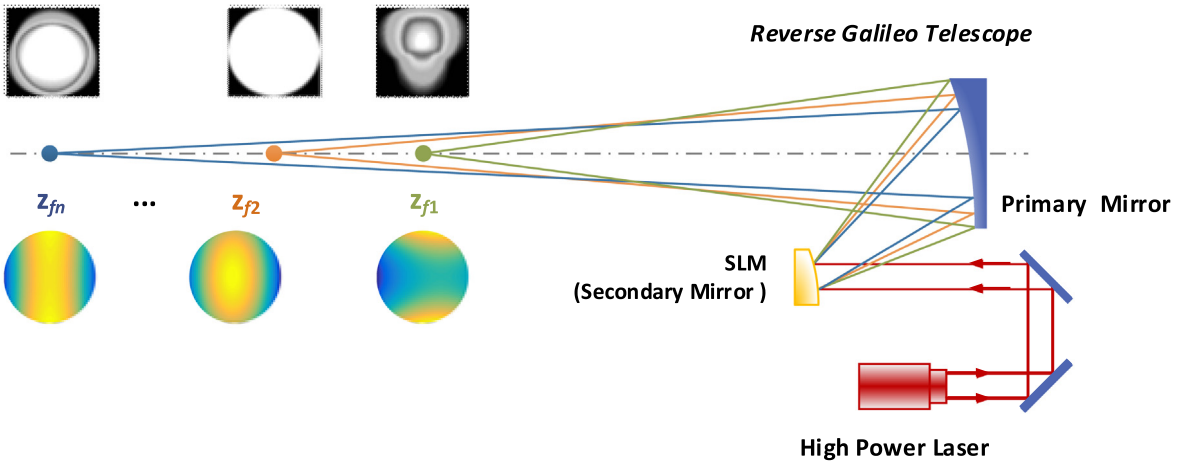


Fig. 2. Optical principle of the RLFS-SLM.

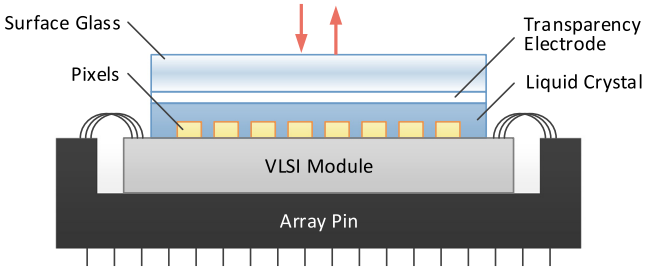


Fig. 3. Structure of the LC-SLM.

second mirror (SLM), thereby optimizing the system design. The wavefront of the secondary mirror (SLM) could be described by Zernike.

The system is monochromatic, so the main consideration is axial spherical aberration, astigmatism and coma. Using the multiple structure of the software, setting the change of object distance can produce the aberration with the change of object distance. According to these aberrations, the system aberrations are corrected by the secondary mirror type in the software optimization.

As shown in Fig. 4, our system has a focal length of 2400 mm with the F/6.8. The plots of modulation transfer functions (MTFs) and pupil maps on the surface of the SLM for 0.5 km operating distance are shown in Fig. 5. At other design distances, the results of MTF are not much different, and MTF has reached the diffraction limit. The MTF plots were close to the diffraction limit after the aberration correction. The pupil maps also demonstrate that the aberration of the different operating distances has been well corrected.

Table 1
The focal lengths of the secondary mirror (SLM) at different operating distances.

Operating distances (km)		The focal length of the SLM (mm)
z_{f1}	0.50	109.5808
z_{f2}	0.75	104.6738
z_{f3}	1.00	103.6662
z_{f4}	1.25	102.8981
z_{f5}	1.50	102.6396

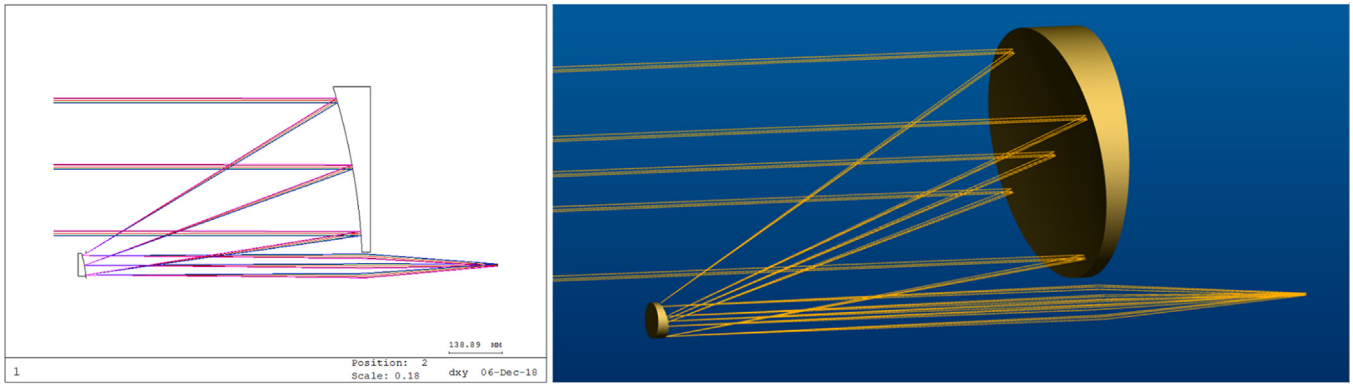


Fig. 4. The optic structure of the RLFS-SLM.

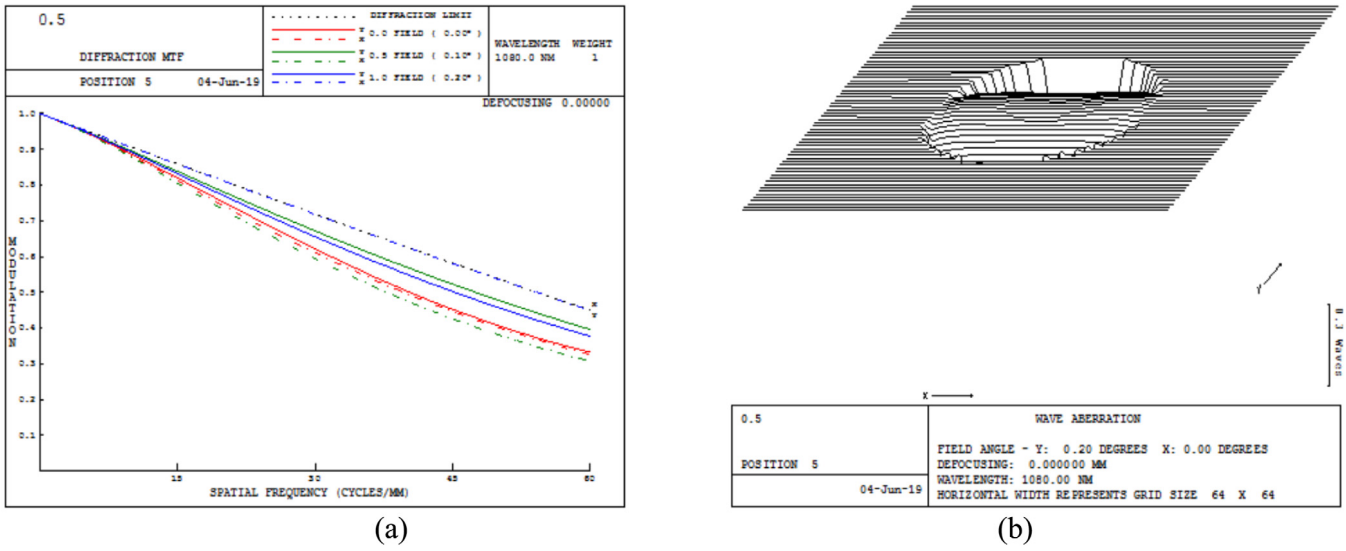


Fig. 5. MTF plots (a) and pupil map (b) on the surface of the SLM (operating distance: 0.5 km).

Table 2
Comparison of aberration and Strehl ratio between RLFSs with or without SLM.

Operating distance (km)	Mech.		SLM	
	RMS (λ)	Strehl ratio	RMS (λ)	Strehl ratio
0.5	0.1487	0.402	0.0672	0.837
0.75	0.1420	0.451	0.0909	0.722
1	0.0217	0.982	0.0987	0.685
1.25	0.1051	0.647	0.1028	0.659
1.5	0.1237	0.502	0.1005	0.647

3.3. Aberration analysis

The RLFS-SLM was compared with the traditional mechanical RLFS (abbreviated as Mech. in Table 2) in both aberration and Strehl ratio. The comparison shows that, after the addition of the SLM, the aberration was reduced while the Strehl ratio was increased. Across the operation range of the RLFS-SLM, the root mean square (RMS) was decreased by 0.016λ on average, and the Strehl ratio saw an average decline of 0.112. Hence, the RLFS-SLM can achieve global optimization.

3.4. Simulation and analysis of the focused spot

The wavefront fitting of the Zernike polynomial coefficients (Section 3.2) was performed on numerical analysis software. The wavefront phase distributions of the secondary mirror (SLM), and the conjugate phases were determined by the SLM adjustment software. The remainder of the 2π modulus was converted into a gray value $\in$ [0, 255] using a phase diagram. Once the phase diagram was loaded onto the LC-SLM, the wavefront could be corrected through phase modulation. The SLM resolution selected in this paper is 1920×1152. The simulated surfaces of the SLM are presented in Fig. 6. Each shade of gray represents the OPD or phase introduced by the SLM at a specific distance.

Next, the convergence of focused spots at different operating distances was analyzed on LightTools, a 3D optical engineering and design software product, for both the RLFS-SLM and the traditional mechanical RLFS. The waist of laser beam inputted to the software was located at the exit of the light chamber. The simulated focused spots at different distances are shown in Fig. 7.

The simulation results in Table 3 show that the focused spots of our RLFS-SLM on average have a 9.5 mm smaller diameter and 54.5 W/cm^2 higher intensity than those of the traditional mechanical RLFS. The results demonstrate that the SLM can improve the focusing ability of the RLFS.

4. Experiments

4.1. Experimental setup

As shown in Fig. 8, the experimental system consists of a high-energy fiber laser, a secondary mirror (SLM), a primary mirror, a diffuse reflector, a camera, a filter, and a monitor. The laser was focused at far

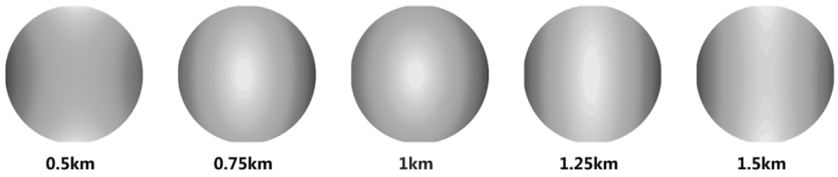


Fig. 6. Wavefront phase distributions on the SLM surface.

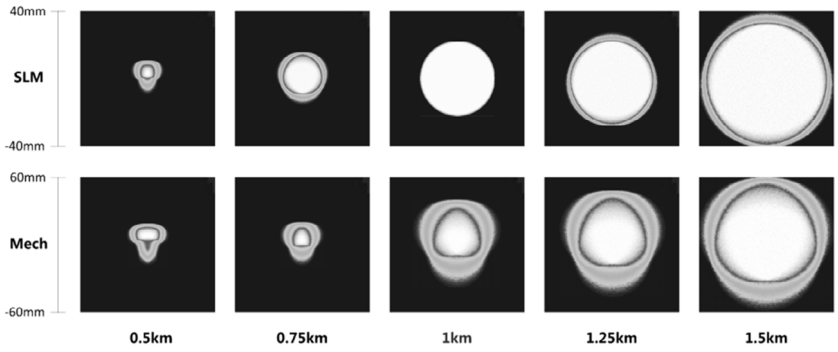


Fig. 7. Simulated focused spots in the RLFS-SLM (first line) and the traditional mechanical RLFS (second line).

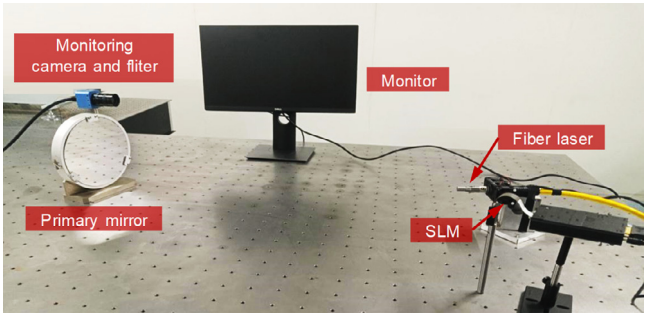


Fig. 8. Setup of the experimental RLFS-SLM.

Table 3
Simulated results of the focused spots in the RLFS-SLM and the traditional mechanical RLFS.

Operating distance (km)	Designed spot diameter (mm)		Designed spot intensity (W/cm^2)	
	SLM	Mech.	SLM	Mech.
0.5	16.9	19.5	446	335
0.75	21.5	25.5	276	196
1	36.4	41.6	96	74
1.25	40.6	53.0	77	45
1.5	50.7	74.1	50	23

field by the RLFS. The camera, which has a long focal length lens, forms a spot monitoring system with the filter. This system was placed next to the primary mirror to capture the image of the focused spot on the surface of diffuse reflector.

The reverse Galileo telescopic system encompasses an off-axis parabolic primary mirror (diameter: 200 mm) and a secondary mirror (SLM) (diameter: 30 mm). The axial distance between the primary and secondary mirrors was 853 mm. The high-energy fiber laser (output power: 1 kW) was placed 100 mm from the secondary mirror (SLM).

The optical path of the experimental system is explained in Fig. 9. First, a laser beam was irradiated onto the SLM. Then, the incident laser beam was modulated by the surface phase of the SLM (Fig. 6). The modulated beam was reflected by the parabolic primary mirror, forming an approximate collimating beam focusing on a remote position. Serving as the operating target, the diffuse reflector reflected the

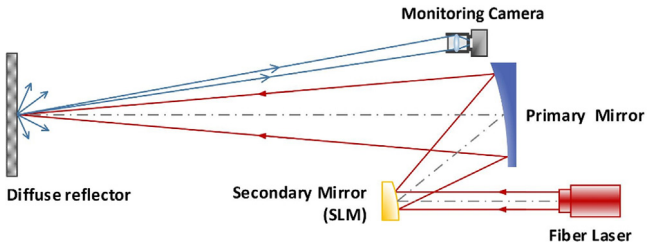


Fig. 9. The optical path of the experimental system.

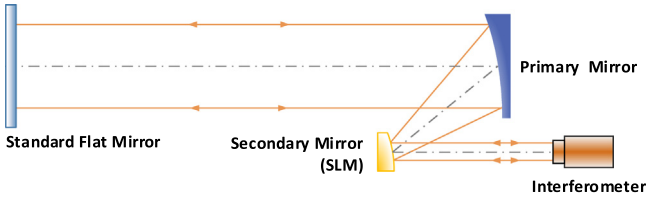


Fig. 10. Zero setting of the optical path.

focused beam. After that, the focused spot could be observed by the monitoring system. The SLM and the primary mirror were protected with laser anti-damage coatings.

4.2. Experimental procedure

The experiment was carried out in five steps:

Step 1. Zero setting (Fig. 10): Use a laser interferometer to create interference fringes of the reflection from the standard flat mirror. Then, tip and tilt the primary mirror and the secondary mirror (SLM) to obtain the expected interference wavefront. During the adjustment, maintain the SLM at the designed phase at infinity.

Step 2. System initialization: Replace the interferometer and standard flat mirror with a high-energy fiber laser and a diffuse reflector, respectively.

Step 3. Experiment of one focused spot: Load one phase diagram onto the SLM (e.g. the spot focusing at the distance of 0.5 km), and light the laser such that the beam is focused by the principle in Fig. 9. Then, tip-tilt the camera, and monitor the centroid of the incident laser

Table 4
Experimental results.

Operating distance (km)	Focused spot diameter (mm)		Focused spot intensity (W/cm^2)	
	SLM	Mech.	SLM	Mech.
0.5	26	39	245	108
0.75	40	42	195	94
1	42	68	61	36
1.25	62	94	43	19
1.5	78	117	27	12

beam on the camera. In addition, place an attenuator in front of the camera to prevent large laser energy from saturating the camera.

Step 4. Experiments of the other focused spots: Load each of the other phase diagrams onto the SLM in turn. Capture the image of each focused spot by the camera.

Step 5. Contrast experiment: Carry out a contrast experiment with the traditional mechanical RLFS. Note that the secondary mirror was also protected with the laser anti-damage coatings, and had almost the same reflectivity as the SLM.

4.3. Experimental results

The results of the above experiments are listed in Table 4. The following conclusions can be drawn from the results:

Conclusion 1: In both the RLFS-SLM and traditional mechanical RLFS, the diameter and intensity of each focused spot both increased with the operating distance. Because the diffraction effect and geometry effect of light beam should be considered simultaneously. In the case of large object distance, geometric effect plays a major role. Therefore, with the increase of the distance, the spot diameter becomes larger and the energy density decreases.

Conclusion 2: The focused spots of the RLFS-SLM were smaller in diameter and higher in intensity than those of the traditional mechanical RLFS. This is because the speckling is reduced through aberration correction by the SLM, making the energy more concentrated.

Conclusion 3 In both systems, there are deviations between the collected spots and the designed spots (Table 3). The deviations may arise from the following factors:

(1) The light transmission in real-world scenario is much more complex than that under the ideal condition. For the designed spots, the spot size, beam waist position and other parameters were all calculated under the ideal condition, without considering the thermal and nonlinear effects of actual laser beams during the transmission.

(2) It is inevitable to have a slight misalignment between the optical axis of the monitoring camera and the normal line of the diffuse reflector.

(3) The focused spots may be enlarged by the assembly error of the experimental device.

(4) The mechanical structure may have intrinsic errors.

Conclusion 4: For the RLFS-SLM, the diameters of the focused spot deviated from the designed values by a root-mean-square error (RMSE) of 6.83 mm. For the traditional mechanical RLFS, this deviation was 12.8 mm, as shown in Fig. 11. Hence, the proposed RLFS-SLM is obviously superior to the traditional mechanical RLFS.

5. Conclusions

This paper puts forward an RLFS-SLM through theoretical analysis and implementation. In this system, an LD-SLM serves as an active secondary mirror of the reverse Galileo telescope system. The beam transmission is controlled by modulating the wavefront of laser. According to simulation and experimental results, the addition of the SLM reduced the diameter of the focused spot by 19%~28%, and enhanced the energy intensity by 58%~110%, from the levels of the traditional mechanical RLFS.

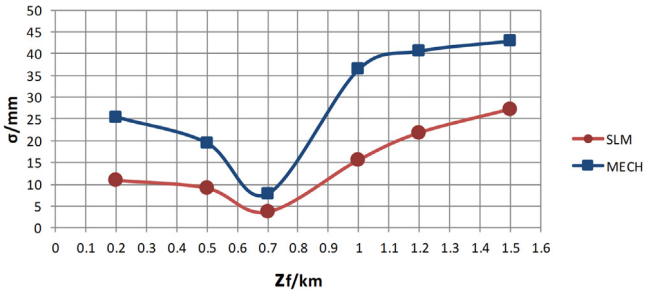


Fig. 11. Spot diameter deviations between the experimental values and the designed values.

In fact, the anti laser damage characteristics of SLM must be considered in the laser damage system. The laser power used in this paper is 1 kW, coated with dielectric film, which can withstand strong laser radiation. In other SLM based high power laser systems, anti laser radiation must be verified in advance.

Our system mainly utilizes off-the-shelf products and common devices, making it cost-effective and flexible. Furthermore, the effectiveness of laser focusing is guaranteed through aberration correction for the SLM wavefront. With these advantages, our system is a promising technique for high-energy laser applications.

To further improve the accuracy, the future research will explore the SLM design for more operating positions and fit the Zernike polynomials of the SLM across the operating range. Besides, the authors will try to quantify the errors induced by system assembly, thermal, nonlinear or other effects related to the imperfectness of the proposed system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] C. Pelosi, D. Fodaro, L. Sforzini, A.L. Monaco, Laser cleaning experiences on sculptures' materials: terracotta, plaster, wood, and wax, in: Fundamentals of Laser-Assisted Micro-and Nanotechnologies 2013, Vol. 9065, 2013, p. 90650X.
[2] B. Rauh, S. Kreling, M. Kolb, M. Geistbeck, S. Boujenfa, M. Suess, K. Dilger, UV-laser cleaning and surface characterization of an aerospace carbon fibre reinforced polymer, Int. J. Adhes. Adhes. 82 (2018) 50–59.
[3] J.S. Shin, S.Y. Oh, H. Park, T.S. Kim, L. Lee, C.M. Chung, J. Lee, Underwater cutting of 50 and 60 mm thick stainless steel plates using a 6-kW fiber laser for dismantling nuclear facilities, Opt. Laser Technol. 115 (2019) 1–8.
[4] K. Tei, D. Sugimoto, M. Endo, S. Takeda, T. Fujioka, Chemical oxygen-iodine laser for decommissioning and dismantlement of nuclear facilities, in: High-Power Lasers in Civil Engineering and Architecture, Vol. 3887, 2000, pp. 162–169.
[5] P. Lutzmann, B. Göhler, C.A. Hill, F.D. van Putten, Laser vibration sensing at fraunhofer IOSB: review and applications, Opt. Eng. 56 (3) (2016) 031215.
[6] F. Di Teodoro, P.M. Belden, P. Ionov, N. Werner, G. Fathi, Development of pulsed fiber lasers for long-range remote sensing, Opt. Eng. 53 (3) (2014) 036105.
[7] Q. Wu, Y.T. Teng, B. Zhang, L.L. Xing, An application of genetic algorithm to improving measurement accuracy of laser interference absolute gravimeter, Instrum. Mes. Métrol. 17 (3) (2018) 467–477.
[8] A. Allen, S.M. Angel, Miniature spatial heterodyne spectrometer for remote laser induced breakdown and Raman spectroscopy using fresnel collection optics, Spectrochim. Acta Part B 149 (2018) 91–98.
[9] X. Zhang, F. Fan, M. Gheisari, G. Srivastava, A novel auto-focus method for image processing using laser triangulation, IEEE Access 7 (2019) 64837–64843.
[10] N. Savage, Digital spatial light modulators, Nat. Photonics 3 (3) (2009) 170.
[11] C. Rosales-Guzmán, A. Forbes, How to shape light with spatial light modulators, 2017, p. 57.
[12] F. Peng, Y.H. Lee, Z. Luo, S.T. Wu, Low voltage blue phase liquid crystal for spatial light modulators, Opt. Lett. 40 (21) (2015) 5097–5100.
[13] T.A. Bartlett, W.C. McDonald, J.N. Hall, Adapting texas instruments DLP technology to demonstrate a phase spatial light modulator, in: Emerging Digital Micromirror Device Based Systems and Applications XI, Vol. 10932, 2019, 109320S.

[14] Z. Kuang, E. Lyon, H. Cheng, V. Page, T. Shenton, G. Dearden, Multi-location laser ignition using a spatial light modulator towards improving automotive gasoline engine performance, Opt. Lasers Eng. 90 (2017) 275–283.
[15] T.W. Clark, R.F. Offer, S. Franke-Arnold, A.S. Arnold, N. Radwell, Comparison of beam generation techniques using a phase only spatial light modulator, Opt. Express 24 (6) (2016) 6249–6264.

Lin Liu received a BS in The measurement and control technology and instrument, MS in Physical Electronics, and MS in optics from Beijing Institute of Technology, China Academic of Electronics and Information Technology, and Beijing Institute of Technology. Her research focuses on laser optical design.

Lin Li received his MS and PhD degrees in optics from Beijing Institute of Technology. His research focuses on optical design. His current position is professor at the Beijing Institute of Technology where his interests are thermal optical, Stray light, optical fabrication and testing, and optical system engineering.

Zhu Zhao received his MS degree in measurement and control technology and instrument, and PhD degree in optics from Beijing Institute of Technology. His research focuses on developing advanced technologies for testing large optics.

Yue Wang received his MS degree in optics from Jilin University. His research focuses on laser system.